Complete 333-variable quartic
Explicit quadratic residuals

ProveIt research artifact

24 October 2026

The single equation

Every unknown ranges over $\mathbb{N} = \{0, 1, 2, \ldots\}$, in the exact order

$$(z_0, z_1, \ldots, z_{332}).$$

The complete equation is

$$\boxed{P_{333}(z_0, \ldots, z_{332}) = \sum_{i=0}^{263} R_i(z_0, \ldots, z_{332})^2 = 0.}$$

333 variables; 264 quadratic residuals; degree 4. The collected polynomial has 4,357 monomials and coefficient height 209,508,384.

All R_i are printed below, without omitted coefficients or implicit auxiliary variables. The symbols R_i are polynomial abbreviations and are not additional unknowns. The index list is zero based and agrees with the supplied JSON. Original coordinate names and kinds are supplied in `variables-333.tsv`; residual names and term counts are supplied in `residuals-333.tsv`.

The report accompanying this equation proves its natural unsolvability equivalent to Con(ZFC), using the explicitly identified inherited C222 calibration. This typesetting file records the entire finite polynomial; it does not assert a new proof-assistant formalization.

Source: `equations/natural333/natural/equation.json`.
SHA-256: `6a32953f56bfceb594c3ea99e011de615d1a66661624a3324207d2c619b372ad`.

Complete residual list

$$R_0 = -144 - 48\,z_0 - 4\,z_0^2 + z_3$$

$$R_1 = -2\,z_2\,z_3 + z_4$$

$$R_2 = -2\,z_2 - 2\,z_2\,z_6 - 2\,z_2\,z_7 + z_5 + z_7$$

$$R_3 = -26 - 4\,z_0 - 4\,z_0\,z_2 - 4\,z_0\,z_8 - 26\,z_2 - 26\,z_8 + z_{11}$$

$$R_4 = 1 + 2\,z_1 - 2\,z_2 - 2\,z_2\,z_9 - 2\,z_8\,z_9 - 3\,z_9$$

$$R_5 = 1 + 2\,z_1 - z_{10} - z_{10}\,z_{11}$$

$$R_6 = 3 + 4\,z_2^2 + 2\,z_{20} - z_{21}\,z_{23}$$

$$\begin{aligned} R_7 = & -20 - 18\,z_{12} - 4\,z_{12}^2 - 8\,z_{12}\,z_{13} - 18\,z_{13} \\ & - 4\,z_{13}^2 + z_{24} \end{aligned}$$

$$\begin{aligned} R_8 = & -156 - 200\,z_{13} - 64\,z_{13}^2 - 32\,z_{13}\,z_{24} - 50\,z_{24} \\ & - 4\,z_{24}^2 + z_{25} \end{aligned}$$

$$\begin{aligned} R_9 = & -42 - 26\,z_{12} - 4\,z_{12}^2 - 8\,z_{12}\,z_{25} - 26\,z_{25} \\ & - 4\,z_{25}^2 + z_{26} \end{aligned}$$

$$\begin{aligned} R_{10} = & -380 - 312\,z_{25} - 64\,z_{25}^2 - 32\,z_{25}\,z_{26} - 78\,z_{26} \\ & - 4\,z_{26}^2 + z_{27} \end{aligned}$$

$$\begin{aligned} R_{11} = & -20 - 18\,z_{13} - 4\,z_{13}^2 - 8\,z_{13}\,z_{14} - 18\,z_{14} \\ & - 4\,z_{14}^2 + z_{28} \end{aligned}$$

$$\begin{aligned} R_{12} = & -20 - 18\,z_{12} - 4\,z_{12}^2 - 8\,z_{12}\,z_{14} - 18\,z_{14} \\ & - 4\,z_{14}^2 + z_{29} \end{aligned}$$

$$\begin{aligned} R_{13} = & -1134 - 84\,z_{12} - 8\,z_{12}^2 - 64\,z_{12}\,z_{14} - 16\,z_{12}\,z_{28} \\ & - 656\,z_{13} - 128\,z_{13}^2 - 256\,z_{13}\,z_{14} - 64\,z_{13}\,z_{24} - 64\,z_{13}\,z_{29} \\ & - 1056\,z_{14} - 256\,z_{14}^2 - 64\,z_{14}\,z_{24} - 64\,z_{14}\,z_{28} - 64\,z_{14}\,z_{29} \\ & - 164\,z_{24} - 8\,z_{24}^2 - 16\,z_{24}\,z_{29} - 92\,z_{28} - 8\,z_{28}^2 \\ & - 172\,z_{29} - 8\,z_{29}^2 + z_{30} \end{aligned}$$

$$\begin{aligned} R_{14} = & -3520 - 2624\,z_{13} - 512\,z_{13}^2 - 1024\,z_{13}\,z_{14} - 256\,z_{13}\,z_{24} \\ & - 256\,z_{13}\,z_{29} - 2752\,z_{14} - 512\,z_{14}^2 - 256\,z_{14}\,z_{24} - 256\,z_{14}\,z_{29} \\ & - 656\,z_{24} - 32\,z_{24}^2 - 64\,z_{24}\,z_{29} - 688\,z_{29} - 32\,z_{29}^2 \\ & - 2\,z_{30} - 2\,z_{30}^2 + z_{31} \end{aligned}$$

$$\begin{aligned} R_{15} = & -731 - 592\,z_{12} - 128\,z_{12}^2 - 256\,z_{12}\,z_{13} - 64\,z_{12}\,z_{28} \\ & - 64\,z_{12}\,z_{29} - 632\,z_{13} - 128\,z_{13}^2 - 64\,z_{13}\,z_{28} - 64\,z_{13}\,z_{29} \\ & - 2\,z_{24} - 156\,z_{28} - 8\,z_{28}^2 - 16\,z_{28}\,z_{29} - 148\,z_{29} \\ & - 8\,z_{29}^2 + z_{32} \end{aligned}$$

$$\begin{aligned} R_{16} = & -2880 - 2368\,z_{12} - 512\,z_{12}^2 - 1024\,z_{12}\,z_{13} - 256\,z_{12}\,z_{28} \\ & - 256\,z_{12}\,z_{29} - 2496\,z_{13} - 512\,z_{13}^2 - 256\,z_{13}\,z_{28} - 256\,z_{13}\,z_{29} \\ & - 624\,z_{28} - 32\,z_{28}^2 - 64\,z_{28}\,z_{29} - 592\,z_{29} - 32\,z_{29}^2 \\ & - 2\,z_{32} - 2\,z_{32}^2 + z_{33} \end{aligned}$$

$$\begin{aligned} R_{17} = & -708 - 100\,z_{12} - 8\,z_{12}^2 - 64\,z_{12}\,z_{14} - 16\,z_{12}\,z_{28} \\ & - 100\,z_{13} - 8\,z_{13}^2 - 64\,z_{13}\,z_{14} - 16\,z_{13}\,z_{29} - 864\,z_{14} \\ & - 256\,z_{14}^2 - 64\,z_{14}\,z_{28} - 64\,z_{14}\,z_{29} - 108\,z_{28} - 8\,z_{28}^2 \\ & - 108\,z_{29} - 8\,z_{29}^2 + z_{34} \end{aligned}$$

$$
\begin{aligned}
R_{18} = & -1416 - 400\,z_{13} - 32\,z_{13}^2 - 256\,z_{13}\,z_{14} - 64\,z_{13}\,z_{29} \\
& - 1728\,z_{14} - 512\,z_{14}^2 - 256\,z_{14}\,z_{29} - 432\,z_{29} - 32\,z_{29}^2 \\
& - 2\,z_{34} - 2\,z_{34}^2 + z_{35}
\end{aligned}
$$

$$
\begin{aligned}
R_{19} = & -731 - 8\,z_{12} - 1216\,z_{13} - 512\,z_{13}^2 - 128\,z_{13}\,z_{24} \\
& - 128\,z_{13}\,z_{28} - 156\,z_{24} - 8\,z_{24}^2 - 16\,z_{24}\,z_{28} - 148\,z_{28} \\
& - 8\,z_{28}^2 - 2\,z_{29} + z_{36}
\end{aligned}
$$

$$
\begin{aligned}
R_{20} = & -2880 - 4864\,z_{13} - 2048\,z_{13}^2 - 512\,z_{13}\,z_{24} - 512\,z_{13}\,z_{28} \\
& - 624\,z_{24} - 32\,z_{24}^2 - 64\,z_{24}\,z_{28} - 592\,z_{28} - 32\,z_{28}^2 \\
& - 2\,z_{36} - 2\,z_{36}^2 + z_{37}
\end{aligned}
$$

$$
\begin{aligned}
R_{21} = & -132 - 184\,z_{13} - 64\,z_{13}^2 - 32\,z_{13}\,z_{24} - 46\,z_{24} \\
& - 4\,z_{24}^2 + z_{38}
\end{aligned}
$$

$$
\begin{aligned}
R_{22} = & -42 - 26\,z_{12} - 4\,z_{12}^2 - 8\,z_{12}\,z_{38} - 26\,z_{38} \\
& - 4\,z_{38}^2 + z_{39}
\end{aligned}
$$

$$
\begin{aligned}
R_{23} = & -380 - 312\,z_{38} - 64\,z_{38}^2 - 32\,z_{38}\,z_{39} - 78\,z_{39} \\
& - 4\,z_{39}^2 + z_{40}
\end{aligned}
$$

$$
\begin{aligned}
R_{24} = & -42 - 26\,z_{13} - 4\,z_{13}^2 - 8\,z_{13}\,z_{40} - 26\,z_{40} \\
& - 4\,z_{40}^2 + z_{41}
\end{aligned}
$$

$$
\begin{aligned}
R_{25} = & -897 - 1352\,z_{13} - 512\,z_{13}^2 - 256\,z_{13}\,z_{24} - 336\,z_{24} \\
& - 32\,z_{24}^2 - 2\,z_{28} + z_{42}
\end{aligned}
$$

$$
\begin{aligned}
R_{26} = & -3544 - 5376\,z_{13} - 2048\,z_{13}^2 - 1024\,z_{13}\,z_{24} - 1344\,z_{24} \\
& - 128\,z_{24}^2 - 2\,z_{42} - 2\,z_{42}^2 + z_{43}
\end{aligned}
$$

$$
\begin{aligned}
R_{27} = & -132 - 184\,z_{12} - 64\,z_{12}^2 - 32\,z_{12}\,z_{29} - 16\,z_{12}\,z_{43} \\
& - 46\,z_{29} - 4\,z_{29}^2 - 4\,z_{29}\,z_{43} - 23\,z_{43} - z_{43}^2 \\
& + z_{44}
\end{aligned}
$$

$$
\begin{aligned}
R_{28} = & -20 - 18\,z_{14} - 4\,z_{14}^2 - 8\,z_{14}\,z_{15} - 18\,z_{15} \\
& - 4\,z_{15}^2 + z_{45}
\end{aligned}
$$

$$
\begin{aligned}
R_{29} = & -804 - 92\,z_{12} - 8\,z_{12}^2 - 64\,z_{12}\,z_{15} - 16\,z_{12}\,z_{45} \\
& - 1088\,z_{14} - 512\,z_{14}^2 - 128\,z_{14}\,z_{28} - 128\,z_{14}\,z_{29} - 336\,z_{15} \\
& - 128\,z_{15}^2 - 64\,z_{15}\,z_{45} - 140\,z_{28} - 8\,z_{28}^2 - 16\,z_{28}\,z_{29} \\
& - 132\,z_{29} - 8\,z_{29}^2 - 84\,z_{45} - 8\,z_{45}^2 + z_{46}
\end{aligned}
$$

$$
\begin{aligned}
R_{30} = & -2304 - 4352\,z_{14} - 2048\,z_{14}^2 - 512\,z_{14}\,z_{28} - 512\,z_{14}\,z_{29} \\
& - 560\,z_{28} - 32\,z_{28}^2 - 64\,z_{28}\,z_{29} - 528\,z_{29} - 32\,z_{29}^2 \\
& - 2\,z_{46} - 2\,z_{46}^2 + z_{47}
\end{aligned}
$$

$$
\begin{aligned}
R_{31} = & -722 - 592\,z_{12} - 128\,z_{12}^2 - 256\,z_{12}\,z_{13} - 64\,z_{12}\,z_{28} \\
& - 64\,z_{12}\,z_{29} - 624\,z_{13} - 128\,z_{13}^2 - 64\,z_{13}\,z_{28} - 64\,z_{13}\,z_{29} \\
& - 2\,z_{14} - 156\,z_{28} - 8\,z_{28}^2 - 16\,z_{28}\,z_{29} - 148\,z_{29} \\
& - 8\,z_{29}^2 + z_{48}
\end{aligned}
$$

$$
\begin{aligned}
R_{32} = & -110 - 168\,z_{14} - 64\,z_{14}^2 - 32\,z_{14}\,z_{28} - 42\,z_{28} \\
& - 4\,z_{28}^2 + z_{49}
\end{aligned}
$$

$$
\begin{aligned}
R_{33} = & -42 - 26\,z_{14} - 4\,z_{14}^2 - 8\,z_{14}\,z_{49} - 26\,z_{49} \\
& - 4\,z_{49}^2 + z_{50}
\end{aligned}
$$

$$
\begin{aligned}
R_{34} = & -342 - 74\,z_{13} - 4\,z_{13}^2 - 32\,z_{13}\,z_{49} - 8\,z_{13}\,z_{50} \\
& - 296\,z_{49} - 64\,z_{49}^2 - 32\,z_{49}\,z_{50} - 74\,z_{50} - 4\,z_{50}^2 \\
& + z_{51}
\end{aligned}
$$

$$
\begin{aligned}
R_{35} = & -4556 - 4320\,z_{49} - 1024\,z_{49}^2 - 512\,z_{49}\,z_{50} - 128\,z_{49}\,z_{51} \\
& - 1080\,z_{50} - 64\,z_{50}^2 - 32\,z_{50}\,z_{51} - 270\,z_{51} - 4\,z_{51}^2 \\
& + z_{52}
\end{aligned}
$$

$$
\begin{aligned}
R_{36} = & -42 - 26\,z_{14} - 4\,z_{14}^2 - 8\,z_{14}\,z_{52} - 26\,z_{52} \\
& - 4\,z_{52}^2 + z_{53}
\end{aligned}
$$

$$
\begin{aligned}
R_{37} = & -342 - 74\,z_{13} - 4\,z_{13}^2 - 32\,z_{13}\,z_{52} - 8\,z_{13}\,z_{53} \\
& - 296\,z_{52} - 64\,z_{52}^2 - 32\,z_{52}\,z_{53} - 74\,z_{53} - 4\,z_{53}^2 \\
& + z_{54}
\end{aligned}
$$

$$
\begin{aligned}
R_{38} = & -2948 - 2368\,z_{12} - 512\,z_{12}^2 - 1024\,z_{12}\,z_{13} - 256\,z_{12}\,z_{28} \\
& - 256\,z_{12}\,z_{29} - 2496\,z_{13} - 512\,z_{13}^2 - 256\,z_{13}\,z_{28} - 256\,z_{13}\,z_{29} \\
& - 624\,z_{28} - 32\,z_{28}^2 - 64\,z_{28}\,z_{29} - 592\,z_{29} - 32\,z_{29}^2 \\
& - 2\,z_{48} - 2\,z_{48}^2 - 32\,z_{52} - 8\,z_{53} - 2\,z_{54} \\
& + z_{55}
\end{aligned}
$$

$$
\begin{aligned}
R_{39} = & -264 - 128\,z_{52} - 32\,z_{53} - 8\,z_{54} - 2\,z_{55} \\
& - 2\,z_{55}^2 + z_{56}
\end{aligned}
$$

$$
\begin{aligned}
R_{40} = & -6 - 10\,z_{12} - 4\,z_{12}^2 - 4\,z_{12}\,z_{56} - 5\,z_{56} \\
& - z_{56}^2 + z_{57}
\end{aligned}
$$

$$
\begin{aligned}
R_{41} = & -12 - 28\,z_{56} - 16\,z_{56}^2 - 16\,z_{56}\,z_{57} - 14\,z_{57} \\
& - 4\,z_{57}^2 + z_{58}
\end{aligned}
$$

$$
\begin{aligned}
R_{42} = & -42 - 26\,z_{14} - 4\,z_{14}^2 - 8\,z_{14}\,z_{58} - 26\,z_{58} \\
& - 4\,z_{58}^2 + z_{59}
\end{aligned}
$$

$$
\begin{aligned}
R_{43} = & -342 - 74\,z_{12} - 4\,z_{12}^2 - 32\,z_{12}\,z_{58} - 8\,z_{12}\,z_{59} \\
& - 296\,z_{58} - 64\,z_{58}^2 - 32\,z_{58}\,z_{59} - 74\,z_{59} - 4\,z_{59}^2 \\
& + z_{60}
\end{aligned}
$$

$$
\begin{aligned}
R_{44} = & -4556 - 4320\,z_{58} - 1024\,z_{58}^2 - 512\,z_{58}\,z_{59} - 128\,z_{58}\,z_{60} \\
& - 1080\,z_{59} - 64\,z_{59}^2 - 32\,z_{59}\,z_{60} - 270\,z_{60} - 4\,z_{60}^2 \\
& + z_{61}
\end{aligned}
$$

$$
\begin{aligned}
R_{45} = & -182 - 216\,z_{14} - 64\,z_{14}^2 - 32\,z_{14}\,z_{28} - 32\,z_{14}\,z_{61} \\
& - 54\,z_{28} - 4\,z_{28}^2 - 8\,z_{28}\,z_{61} - 54\,z_{61} - 4\,z_{61}^2 \\
& + z_{62}
\end{aligned}
$$

$$
\begin{aligned}
R_{46} = & -4556 - 4320\,z_{52} - 1024\,z_{52}^2 - 512\,z_{52}\,z_{53} - 128\,z_{52}\,z_{54} \\
& - 1080\,z_{53} - 64\,z_{53}^2 - 32\,z_{53}\,z_{54} - 270\,z_{54} - 4\,z_{54}^2 \\
& + z_{63}
\end{aligned}
$$

$$
\begin{aligned}
R_{47} = & -754 - 148\,z_{13} - 8\,z_{13}^2 - 64\,z_{13}\,z_{61} - 16\,z_{13}\,z_{62} \\
& - 624\,z_{61} - 128\,z_{61}^2 - 64\,z_{61}\,z_{62} - 156\,z_{62} - 8\,z_{62}^2 \\
& - 2\,z_{63} + z_{64}
\end{aligned}
$$

$$
R_{48} = -z_{64} - z_{64}^2 + z_{65}
$$

$$
\begin{aligned}
R_{49} = & -132 - 184\,z_{13} - 64\,z_{13}^2 - 32\,z_{13}\,z_{14} - 32\,z_{13}\,z_{24} \\
& - 46\,z_{14} - 4\,z_{14}^2 - 8\,z_{14}\,z_{24} - 46\,z_{24} - 4\,z_{24}^2 \\
& + z_{66}
\end{aligned}
$$

$$
\begin{aligned}
R_{50} = & -5414 - 7728\,z_{13} - 3200\,z_{13}^2 - 1600\,z_{13}\,z_{24} - 320\,z_{13}\,z_{66} \\
& - 148\,z_{14} - 8\,z_{14}^2 - 64\,z_{14}\,z_{63} - 16\,z_{14}\,z_{65} - 1932\,z_{24} \\
& - 200\,z_{24}^2 - 80\,z_{24}\,z_{66} - 624\,z_{63} - 128\,z_{63}^2 - 64\,z_{63}\,z_{65} \\
& - 156\,z_{65} - 8\,z_{65}^2 - 388\,z_{66} - 8\,z_{66}^2 + z_{67}
\end{aligned}
$$

$$
\begin{aligned}
R_{51} = & -3000 - 592\,z_{14} - 32\,z_{14}^2 - 256\,z_{14}\,z_{63} - 64\,z_{14}\,z_{65} \\
& - 2496\,z_{63} - 512\,z_{63}^2 - 256\,z_{63}\,z_{65} - 624\,z_{65} - 32\,z_{65}^2 \\
& - 2\,z_{67} - 2\,z_{67}^2 + z_{68}
\end{aligned}
$$

$$
R_{52} = -2 - 3\,z_{68} - z_{68}^2 + z_{69}
$$

$$
R_{53} = -4\,z_2 - 4\,z_2\,z_{17} + z_{70}
$$

$$
R_{54} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} - 4\,z_{17}^2 + z_{71}
$$

$$
R_{55} = -2\,z_{27}\,z_{70} - 4\,z_{70} + z_{72}
$$

$$
\begin{aligned}
R_{56} = & -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{72} - 4\,z_{17}^2 - 4\,z_{17}\,z_{72} \\
& - z_{72}^2 + z_{73}
\end{aligned}
$$

$$
R_{57} = -z_{31}\,z_{70} + z_{74}
$$

$$
\begin{aligned}
R_{58} = & -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{74} - 4\,z_{17}^2 - 4\,z_{17}\,z_{74} \\
& - z_{74}^2 + z_{75}
\end{aligned}
$$

$$R_{59} = -z_{33}\,z_{70} + z_{76}$$

$$R_{60} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{76} - 4\,z_{17}^2 - 4\,z_{17}\,z_{76} - z_{76}^2 + z_{77}$$

$$R_{61} = -z_{35}\,z_{70} + z_{78}$$

$$R_{62} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{78} - 4\,z_{17}^2 - 4\,z_{17}\,z_{78} - z_{78}^2 + z_{79}$$

$$R_{63} = -z_{37}\,z_{70} + z_{80}$$

$$R_{64} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{80} - 4\,z_{17}^2 - 4\,z_{17}\,z_{80} - z_{80}^2 + z_{81}$$

$$R_{65} = -8\,z_{13}\,z_{70} - 2\,z_{41}\,z_{70} - 8\,z_{70} + z_{82}$$

$$R_{66} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{82} - 4\,z_{17}^2 - 4\,z_{17}\,z_{82} - z_{82}^2 + z_{83}$$

$$R_{67} = -4\,z_{43}\,z_{70} - 2\,z_{44}\,z_{70} + z_{84}$$

$$R_{68} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{84} - 4\,z_{17}^2 - 4\,z_{17}\,z_{84} - z_{84}^2 + z_{85}$$

$$R_{69} = -z_{47}\,z_{70} + z_{86}$$

$$R_{70} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{86} - 4\,z_{17}^2 - 4\,z_{17}\,z_{86} - z_{86}^2 + z_{87}$$

$$R_{71} = -4\,z_{12}^2 + 8\,z_{12}\,z_{14} - 4\,z_{14}^2 + z_{88}$$

$$R_{72} = -4\,z_{13}^2 + 8\,z_{13}\,z_{14} - 4\,z_{14}^2 + z_{89}$$

$$R_{73} = -z_{88}\,z_{89} + z_{90}$$

$$R_{74} = -9 - 12\,z_{15} - 4\,z_{15}^2 + z_{91}$$

$$R_{75} = -z_{90}^2 + 2\,z_{90}\,z_{91} - z_{91}^2 + z_{92}$$

$$R_{76} = -2\,z_{69}\,z_{70} - 4\,z_{70} + z_{93}$$

$$R_{77} = -4\,z_{16}^2 + 8\,z_{16}\,z_{17} + 4\,z_{16}\,z_{93} - 4\,z_{17}^2 - 4\,z_{17}\,z_{93} - z_{93}^2 + z_{94}$$

$$R_{78} = -4\,z_2 - 4\,z_2\,z_{13} + z_{95}$$

$$R_{79} = -2 - 2\,z_{17} - 2\,z_{17}\,z_{95} - 2\,z_{95} + z_{96}$$

$$R_{80} = -4\,z_2 - 4\,z_2\,z_{12} + z_{97}$$

$$R_{81} = -16\,z_2 - 16\,z_2\,z_{13} - 4\,z_2\,z_{24} + z_{98}$$

$$R_{82} = -2 - 2\,z_{16} - 2\,z_{16}\,z_{97} - 2\,z_{97} + z_{99}$$

$$R_{83} = -z_{98}\,z_{99} - z_{99} + z_{100}$$

$$R_{84} = -z_{96}^2 + 2\,z_{96}\,z_{100} - z_{100}^2 + z_{101}$$

$$R_{85} = -20\,z_2 - 16\,z_2\,z_{13} - 4\,z_2\,z_{24} + z_{102}$$

$$R_{86} = -2 - 2\,z_{17} - 2\,z_{17}\,z_{102} - 2\,z_{102} + z_{103}$$

$$R_{87} = -2 - 2\,z_{16} - 2\,z_{16}\,z_{95} - 2\,z_{95} + z_{104}$$

$$R_{88} = -z_{103}^2 + 2\,z_{103}\,z_{104} - z_{104}^2 + z_{105}$$

$$R_{89} = -z_{71}\,z_{73} + z_{106}$$

$$R_{90} = -z_{75}\,z_{106} + z_{107}$$

$$R_{91} = -z_{77}\,z_{107} + z_{108}$$

$$R_{92} = -z_{79}\,z_{108} + z_{109}$$

$$R_{93} = -z_{81}\,z_{109} + z_{110}$$

$$R_{94} = -z_{83}\,z_{110} + z_{111}$$

$$R_{95} = -z_{85}\,z_{111} + z_{112}$$

$$R_{96} = -z_{87}\,z_{112} + z_{113}$$

$$R_{97} = -z_{92}\,z_{113} - z_{94}\,z_{113} + z_{114}$$

$$R_{98} = -z_{101}\,z_{114} + z_{115}$$

$$R_{99} = -6 - 6\,z_2 - 4\,z_2\,z_{20} - 6\,z_8 - 4\,z_8\,z_{20} - 4\,z_{20} + z_{116}$$

$$R_{100} = -2 - 2\,z_{18} - 2\,z_{18}\,z_{116} - 2\,z_{116} + z_{117}$$

$$
\begin{aligned}
R_{101} = & -6 - 4\,z_2 - 4\,z_2\,z_{19} - 4\,z_8 - 4\,z_8\,z_{19} \\
& - 6\,z_{19} - 2\,z_{19}\,z_{116} - 2\,z_{116} + z_{118}
\end{aligned}
$$

$$
\begin{aligned}
R_{102} = & \, 8\,z_1^2 - 8\,z_1\,z_{16} - 8\,z_1\,z_{17} - 4\,z_1\,z_{117} - 4\,z_1\,z_{118} \\
& + 4\,z_{16}^2 + 4\,z_{16}\,z_{117} + 4\,z_{17}^2 + 4\,z_{17}\,z_{118} - z_{22}\,z_{119} \\
& + z_{105}\,z_{115} + z_{117}^2 + z_{118}^2
\end{aligned}
$$

$$
\begin{aligned}
R_{103} = & -18 - 2\,z_{12} - 2\,z_{13} - 2\,z_{14} - 2\,z_{15} \\
& - 2\,z_{16} - 2\,z_{17} - 2\,z_{18} - 2\,z_{19} - 2\,z_{20} \\
& + 18\,z_{22}^2 - z_{120}
\end{aligned}
$$

$$
R_{104} = 2 + z_6 + z_7 - z_{22}\,z_{121} + z_{190}
$$

$$
\begin{aligned}
R_{105} = & -2\,z_{187} - z_{187}^2 - 2\,z_{187}\,z_{188} - 2\,z_{188} - z_{188}^2 \\
& + z_{313} + z_{314}
\end{aligned}
$$

$$
R_{106} = -z_{22} - z_{22}\,z_{280} - z_{22}\,z_{281} + z_{122}
$$

$$
R_{107} = -z_{123} - z_{123}\,z_{126} + z_{127}
$$

$$
R_{108} = -z_{123}\,z_{127} - z_{127} + z_{128}
$$

$$
R_{109} = -1 + z_{124}^2 + z_{127}^2 - z_{128}^2
$$

$$
\begin{aligned}
R_{110} = & -10 - 6\,z_6 - z_6^2 - 2\,z_6\,z_7 + 2\,z_6\,z_{124} \\
& + 2\,z_6\,z_{125} - 2\,z_6\,z_{190} - 6\,z_7 - z_7^2 + 2\,z_7\,z_{124} \\
& + 2\,z_7\,z_{125} - 2\,z_7\,z_{190} - z_{122}\,z_{129} + 6\,z_{124} + 2\,z_{124}\,z_{190} \\
& + 6\,z_{125} + 2\,z_{125}\,z_{190} - 6\,z_{190} - z_{190}^2
\end{aligned}
$$

$$
R_{111} = -z_{124} - z_{124}\,z_{131} - z_{125} - z_{125}\,z_{131} + z_{134}
$$

$$
R_{112} = z_{130}^2 + 2\,z_{131} + z_{131}^2 - z_{134}^2
$$

$$
R_{113} = -4 - 4\,z_{131} - 4\,z_{131}\,z_{132} - 4\,z_{132} + z_{135}
$$

$$
R_{114} = 1 - z_{124} - z_{125} - z_{130}\,z_{133} + z_{135}
$$

$$
R_{115} = -z_{124}\,z_{137} - z_{125}\,z_{137} + z_{140}
$$

$$
R_{116} = -1 + z_{136}^2 + z_{137}^2 - z_{140}^2
$$

$$
R_{117} = -z_{135}\,z_{139} - z_{139} + z_{141}
$$

$$
R_{118} = -1 + z_{138}^2 + z_{139}^2 - z_{141}^2
$$

$$
R_{119} = -z_{130}\,z_{142} - z_{136} + z_{138}
$$

$$R_{120} = -12 - 2\,z_0 - 4\,z_{137}\,z_{143} + z_{139}$$

$$\begin{aligned} R_{121} = & -2 - 2\,z_4 - z_4^2 + 2\,z_4\,z_{124} + 2\,z_4\,z_{125} \\ & + 2\,z_{124} + 2\,z_{125} - z_{144} \end{aligned}$$

$$\begin{aligned} R_{122} = & \, z_4\,z_{137} - z_5 + z_{136} + z_{137} - z_{140} \\ & - z_{144}\,z_{145} \end{aligned}$$

$$R_{123} = -z_{124}\,z_{147} - z_{125}\,z_{147} + z_{150}$$

$$R_{124} = -1 + z_{146}^2 + z_{147}^2 - z_{150}^2$$

$$R_{125} = -z_{135}\,z_{149} - z_{149} + z_{151}$$

$$R_{126} = -1 + z_{148}^2 + z_{149}^2 - z_{151}^2$$

$$R_{127} = -z_{130}\,z_{152} - z_{146} + z_{148}$$

$$\begin{aligned} R_{128} = & -36 - 4\,z_0 - 4\,z_1 - 8\,z_2 - 4\,z_8 \\ & - 4\,z_{147}\,z_{153} + z_{149} \end{aligned}$$

$$R_{129} = -1 - z_{21}^2 + 2\,z_{21}\,z_{124} + 2\,z_{21}\,z_{125} - z_{154}$$

$$R_{130} = z_{21}\,z_{147} - z_{22} + z_{146} - z_{150} - z_{154}\,z_{155}$$

$$\begin{aligned} R_{131} = & -2 - 2\,z_5 - z_5^2 - 2\,z_5\,z_{22} + 2\,z_5\,z_{124} \\ & + 2\,z_5\,z_{125} - 2\,z_{22} - z_{22}^2 + 2\,z_{22}\,z_{124} + 2\,z_{22}\,z_{125} \\ & + 2\,z_{124} + 2\,z_{125} - z_{156} \end{aligned}$$

$$\begin{aligned} R_{132} = & \, z_5\,z_{147} + z_{22}\,z_{147} - z_{121} + z_{146} + z_{147} \\ & - z_{150} - z_{156}\,z_{157} \end{aligned}$$

$$\begin{aligned} R_{133} = & -5 - 4\,z_6 - z_6^2 - 2\,z_6\,z_7 + 2\,z_6\,z_{124} \\ & + 2\,z_6\,z_{125} - 2\,z_6\,z_{190} - 4\,z_7 - z_7^2 + 2\,z_7\,z_{124} \\ & + 2\,z_7\,z_{125} - 2\,z_7\,z_{190} + 4\,z_{124} + 2\,z_{124}\,z_{190} + 4\,z_{125} \\ & + 2\,z_{125}\,z_{190} - z_{158} - 4\,z_{190} - z_{190}^2 \end{aligned}$$

$$\begin{aligned} R_{134} = & -1 + z_6\,z_{137} + z_7\,z_{137} + z_{136} + 2\,z_{137} \\ & + z_{137}\,z_{190} - z_{140} - z_{158}\,z_{159} - z_{187} - z_{188} \end{aligned}$$

$$R_{135} = -z_{124}\,z_{161} - z_{125}\,z_{161} + z_{164}$$

$$R_{136} = -1 + z_{160}^2 + z_{161}^2 - z_{164}^2$$

$$R_{137} = -z_{135}\,z_{163} - z_{163} + z_{165}$$

$$R_{138} = -1 + z_{162}^2 + z_{163}^2 - z_{165}^2$$

$$R_{139} = -z_{130}\,z_{166} - z_{160} + z_{162}$$

$$R_{140} = -2 - 2\,z_2 - 2\,z_8 - 4\,z_{161}\,z_{167} + z_{163}$$

$$\begin{aligned} R_{141} = {} & -1 + z_6\,z_{161} + z_7\,z_{161} - z_{158}\,z_{168} + z_{160} \\ & + 2\,z_{161} + z_{161}\,z_{190} - z_{164} - z_{257} - z_{258} \end{aligned}$$

$$R_{142} = -z_{124}\,z_{170} - z_{125}\,z_{170} + z_{173}$$

$$R_{143} = -1 + z_{169}^2 + z_{170}^2 - z_{173}^2$$

$$R_{144} = -z_{135}\,z_{172} - z_{172} + z_{174}$$

$$R_{145} = -1 + z_{171}^2 + z_{172}^2 - z_{174}^2$$

$$R_{146} = -z_{130}\,z_{175} - z_{169} + z_{171}$$

$$R_{147} = -2 - 2\,z_1 - 4\,z_{170}\,z_{176} + z_{172}$$

$$\begin{aligned} R_{148} = {} & -1 + z_6\,z_{170} + z_7\,z_{170} - z_{158}\,z_{177} + z_{169} \\ & + 2\,z_{170} + z_{170}\,z_{190} - z_{173} - z_{280} - z_{281} \end{aligned}$$

$$R_{149} = -z_{124}\,z_{179} - z_{125}\,z_{179} + z_{182}$$

$$R_{150} = -1 + z_{178}^2 + z_{179}^2 - z_{182}^2$$

$$R_{151} = -z_{135}\,z_{181} - z_{181} + z_{183}$$

$$R_{152} = -1 + z_{180}^2 + z_{181}^2 - z_{183}^2$$

$$R_{153} = -z_{130}\,z_{184} - z_{178} + z_{180}$$

$$R_{154} = -12 - 2\,z_0 - z_4 - 4\,z_{179}\,z_{185} + z_{181}$$

$$\begin{aligned} R_{155} = {} & -2 - z_6 - z_6\,z_{187} - z_6\,z_{188} - z_7 \\ & - z_7\,z_{187} - z_7\,z_{188} - 2\,z_{187} - z_{187}\,z_{190} - 2\,z_{188} \\ & - z_{188}\,z_{190} + z_{189} - z_{190} \end{aligned}$$

$$\begin{aligned} R_{156} = {} & -1 - z_6 + z_6\,z_{179} - z_6\,z_{187} - z_6\,z_{188} \\ & - z_7 + z_7\,z_{179} - z_7\,z_{187} - z_7\,z_{188} + z_{178} \\ & + 3\,z_{179} + z_{179}\,z_{190} - z_{182} - z_{186}\,z_{189} - 2\,z_{187} \\ & - z_{188} \end{aligned}$$

$$R_{157} = -z_{124}\,z_{192} - z_{125}\,z_{192} + z_{195}$$

$$R_{158} = -1 + z_{191}^2 + z_{192}^2 - z_{195}^2$$

$$R_{159} = -z_{135}\, z_{194} - z_{194} + z_{196}$$

$$R_{160} = -1 + z_{193}^2 + z_{194}^2 - z_{196}^2$$

$$R_{161} = -z_{130}\, z_{197} - z_{191} + z_{193}$$

$$R_{162} = -12 - 2\, z_0 - 4\, z_2^2 - 4\, z_{192}\, z_{198} + z_{194}$$

$$\begin{aligned} R_{163} = {} & 1 + z_6 + z_7 - z_{21} + z_{190} \\ & - z_{202} \end{aligned}$$

$$\begin{aligned} R_{164} = {} & z_6\, z_{192} + z_7\, z_{192} - z_{21} - z_{21}\, z_{187} - z_{21}\, z_{188} \\ & - z_{189}\, z_{199} + z_{190}\, z_{192} + z_{191} + 3\, z_{192} - z_{195} \\ & - z_{200} \end{aligned}$$

$$R_{165} = z_{187} + z_{188} - z_{200} - z_{201}$$

$$R_{166} = -z_{124}\, z_{204} - z_{125}\, z_{204} + z_{207}$$

$$R_{167} = -1 + z_{203}^2 + z_{204}^2 - z_{207}^2$$

$$R_{168} = -z_{135}\, z_{206} - z_{206} + z_{208}$$

$$R_{169} = -1 + z_{205}^2 + z_{206}^2 - z_{208}^2$$

$$R_{170} = -z_{130}\, z_{209} - z_{203} + z_{205}$$

$$R_{171} = -2 - 2\, z_{12} - 4\, z_{204}\, z_{210} + z_{206}$$

$$R_{172} = -z_{22} - z_{22}\, z_{187} - z_{22}\, z_{188} + z_{214}$$

$$\begin{aligned} R_{173} = {} & z_6\, z_{204} + z_7\, z_{204} + z_{190}\, z_{204} + z_{203} + 3\, z_{204} \\ & - z_{207} - z_{211}\, z_{214} - z_{212} \end{aligned}$$

$$R_{174} = z_{187} + z_{188} - z_{212} - z_{213}$$

$$R_{175} = -z_{124}\, z_{216} - z_{125}\, z_{216} + z_{219}$$

$$R_{176} = -1 + z_{215}^2 + z_{216}^2 - z_{219}^2$$

$$R_{177} = -z_{135}\, z_{218} - z_{218} + z_{220}$$

$$R_{178} = -1 + z_{217}^2 + z_{218}^2 - z_{220}^2$$

$$R_{179} = -z_{130}\,z_{221} - z_{215} + z_{217}$$

$$R_{180} = -2 - 2\,z_{13} - 4\,z_{216}\,z_{222} + z_{218}$$

$$R_{181} = z_6\,z_{216} + z_7\,z_{216} + z_{190}\,z_{216} - z_{214}\,z_{223} + z_{215} + 3\,z_{216} - z_{219} - z_{224}$$

$$R_{182} = z_{187} + z_{188} - z_{224} - z_{225}$$

$$R_{183} = -z_{124}\,z_{227} - z_{125}\,z_{227} + z_{230}$$

$$R_{184} = -1 + z_{226}^2 + z_{227}^2 - z_{230}^2$$

$$R_{185} = -z_{135}\,z_{229} - z_{229} + z_{231}$$

$$R_{186} = -1 + z_{228}^2 + z_{229}^2 - z_{231}^2$$

$$R_{187} = -z_{130}\,z_{232} - z_{226} + z_{228}$$

$$R_{188} = -2 - 2\,z_{14} - 4\,z_{227}\,z_{233} + z_{229}$$

$$R_{189} = z_6\,z_{227} + z_7\,z_{227} + z_{190}\,z_{227} - z_{214}\,z_{234} + z_{226} + 3\,z_{227} - z_{230} - z_{235}$$

$$R_{190} = z_{187} + z_{188} - z_{235} - z_{236}$$

$$R_{191} = -z_{124}\,z_{238} - z_{125}\,z_{238} + z_{241}$$

$$R_{192} = -1 + z_{237}^2 + z_{238}^2 - z_{241}^2$$

$$R_{193} = -z_{135}\,z_{240} - z_{240} + z_{242}$$

$$R_{194} = -1 + z_{239}^2 + z_{240}^2 - z_{242}^2$$

$$R_{195} = -z_{130}\,z_{243} - z_{237} + z_{239}$$

$$R_{196} = -2 - 2\,z_{15} - 4\,z_{238}\,z_{244} + z_{240}$$

$$R_{197} = z_6\,z_{238} + z_7\,z_{238} + z_{190}\,z_{238} - z_{214}\,z_{245} + z_{237} + 3\,z_{238} - z_{241} - z_{246}$$

$$R_{198} = z_{187} + z_{188} - z_{246} - z_{247}$$

$$R_{199} = -z_{124}\,z_{249} - z_{125}\,z_{249} + z_{252}$$

$$R_{200} = -1 + z_{248}^2 + z_{249}^2 - z_{252}^2$$

$$R_{201} = -z_{135}\,z_{251} - z_{251} + z_{253}$$

$$R_{202} = -1 + z_{250}^2 + z_{251}^2 - z_{253}^2$$

$$R_{203} = -z_{130}\,z_{254} - z_{248} + z_{250}$$

$$R_{204} = -2 - 2\,z_{16} - 4\,z_{249}\,z_{255} + z_{251}$$

$$R_{205} = -z_{22} - z_{22}\,z_{257} - z_{22}\,z_{258} + z_{259}$$

$$R_{206} = z_6\,z_{249} + z_7\,z_{249} + z_{190}\,z_{249} + z_{248} + 3\,z_{249} - z_{252} - z_{256}\,z_{259} - z_{257}$$

$$R_{207} = -z_{124}\,z_{261} - z_{125}\,z_{261} + z_{264}$$

$$R_{208} = -1 + z_{260}^2 + z_{261}^2 - z_{264}^2$$

$$R_{209} = -z_{135}\,z_{263} - z_{263} + z_{265}$$

$$R_{210} = -1 + z_{262}^2 + z_{263}^2 - z_{265}^2$$

$$R_{211} = -z_{130}\,z_{266} - z_{260} + z_{262}$$

$$R_{212} = -2 - 2\,z_{17} - 4\,z_{261}\,z_{267} + z_{263}$$

$$R_{213} = z_6\,z_{261} + z_7\,z_{261} + z_{190}\,z_{261} - z_{259}\,z_{268} + z_{260} + 3\,z_{261} - z_{264} - z_{269}$$

$$R_{214} = z_{257} + z_{258} - z_{269} - z_{270}$$

$$R_{215} = -z_{124}\,z_{272} - z_{125}\,z_{272} + z_{275}$$

$$R_{216} = -1 + z_{271}^2 + z_{272}^2 - z_{275}^2$$

$$R_{217} = -z_{135}\,z_{274} - z_{274} + z_{276}$$

$$R_{218} = -1 + z_{273}^2 + z_{274}^2 - z_{276}^2$$

$$R_{219} = -z_{130}\,z_{277} - z_{271} + z_{273}$$

$$R_{220} = -2 - 2\,z_{18} - 4\,z_{272}\,z_{278} + z_{274}$$

$$R_{221} = z_6\,z_{272} + z_7\,z_{272} - z_{122}\,z_{279} + z_{190}\,z_{272} + z_{271} + 3\,z_{272} - z_{275} - z_{280}$$

$$R_{222} = -z_{124}\,z_{283} - z_{125}\,z_{283} + z_{286}$$

$$R_{223} = -1 + z_{282}^2 + z_{283}^2 - z_{286}^2$$

$$R_{224} = -z_{135}\, z_{285} - z_{285} + z_{287}$$

$$R_{225} = -1 + z_{284}^2 + z_{285}^2 - z_{287}^2$$

$$R_{226} = -z_{130}\, z_{288} - z_{282} + z_{284}$$

$$R_{227} = -2 - 2\, z_{19} - 4\, z_{283}\, z_{289} + z_{285}$$

$$\begin{aligned} R_{228} = {} & z_6\, z_{283} + z_7\, z_{283} - z_{122}\, z_{290} + z_{190}\, z_{283} + z_{282} \\ & + 3\, z_{283} - z_{286} - z_{291} \end{aligned}$$

$$R_{229} = z_{280} + z_{281} - z_{291} - z_{292}$$

$$R_{230} = -z_{124}\, z_{294} - z_{125}\, z_{294} + z_{297}$$

$$R_{231} = -1 + z_{293}^2 + z_{294}^2 - z_{297}^2$$

$$R_{232} = -z_{135}\, z_{296} - z_{296} + z_{298}$$

$$R_{233} = -1 + z_{295}^2 + z_{296}^2 - z_{298}^2$$

$$R_{234} = -z_{130}\, z_{299} - z_{293} + z_{295}$$

$$R_{235} = -2 - 2\, z_{20} - 4\, z_{294}\, z_{300} + z_{296}$$

$$\begin{aligned} R_{236} = {} & z_6\, z_{294} + z_7\, z_{294} + z_{190}\, z_{294} - z_{214}\, z_{301} + z_{293} \\ & + 3\, z_{294} - z_{297} - z_{302} \end{aligned}$$

$$R_{237} = z_{187} + z_{188} - z_{302} - z_{303}$$

$$R_{238} = -z_{124}\, z_{305} - z_{125}\, z_{305} + z_{308}$$

$$R_{239} = -1 + z_{304}^2 + z_{305}^2 - z_{308}^2$$

$$R_{240} = -z_{135}\, z_{307} - z_{307} + z_{309}$$

$$R_{241} = -1 + z_{306}^2 + z_{307}^2 - z_{309}^2$$

$$R_{242} = -z_{130}\, z_{310} - z_{304} + z_{306}$$

$$\begin{aligned} R_{243} = {} & -78 - 4\, z_{12} - 52\, z_{13} - 8\, z_{24} - 4\, z_{27} \\ & - 2\, z_{31} - 2\, z_{33} - 2\, z_{35} - 2\, z_{37} - 4\, z_{41} \\ & - 8\, z_{43} - 4\, z_{44} - 2\, z_{47} - 4\, z_{69} - 4\, z_{305}\, z_{311} \\ & + z_{307} \end{aligned}$$

$$R_{244} = -z_{22} - z_{22}\,z_{313} - z_{22}\,z_{314} + z_{315}$$

$$\begin{aligned} R_{245} = {} & z_6\,z_{305} + z_7\,z_{305} + z_{190}\,z_{305} + z_{304} + 3\,z_{305} \\ & - z_{308} - z_{312}\,z_{315} - z_{313} \end{aligned}$$

$$R_{246} = -z_{137}\,z_{147} + z_{316}$$

$$R_{247} = -z_{161}\,z_{316} + z_{317}$$

$$R_{248} = -z_{170}\,z_{317} + z_{318}$$

$$R_{249} = -z_{179}\,z_{318} + z_{319}$$

$$R_{250} = -z_{192}\,z_{319} + z_{320}$$

$$R_{251} = -z_{204}\,z_{320} + z_{321}$$

$$R_{252} = -z_{216}\,z_{321} + z_{322}$$

$$R_{253} = -z_{227}\,z_{322} + z_{323}$$

$$R_{254} = -z_{238}\,z_{323} + z_{324}$$

$$R_{255} = -z_{249}\,z_{324} + z_{325}$$

$$R_{256} = -z_{261}\,z_{325} + z_{326}$$

$$R_{257} = -z_{272}\,z_{326} + z_{327}$$

$$R_{258} = -z_{283}\,z_{327} + z_{328}$$

$$R_{259} = -z_{294}\,z_{328} + z_{329}$$

$$R_{260} = -z_{305}\,z_{329} + z_{330}$$

$$R_{261} = -z_{330}^2 + z_{331}$$

$$R_{262} = 1 + z_{131} - z_{331} - z_{331}\,z_{332}$$

$$\begin{aligned} R_{263} = {} & -181 - 10\,z_0 - 6\,z_1 - 10\,z_2 - 4\,z_2^2 \\ & - 2\,z_4 - 2\,z_5 - z_6 - z_7 - 6\,z_8 \\ & - 6\,z_{12} - 54\,z_{13} - 2\,z_{14} - 2\,z_{15} - 2\,z_{16} \\ & - 2\,z_{17} - 2\,z_{18} - 2\,z_{19} - 2\,z_{20} - z_{21} \\ & - 2\,z_{22} - 8\,z_{24} - 4\,z_{27} - 2\,z_{31} - 2\,z_{33} \\ & - 2\,z_{35} - 2\,z_{37} - 4\,z_{41} - 8\,z_{43} - 4\,z_{44} \\ & - 2\,z_{47} - 4\,z_{69} - z_{121} + z_{123} - z_{187} \\ & - z_{188} - z_{190} - z_{257} - z_{258} - z_{280} \\ & - z_{281} \end{aligned}$$